

STATISTICS WORKSHOP

From raw data to business decisions

Prepared from class notes + workshop reference material

DATA

SUMMARISE

PROBABILITY

DECIDE

WORKSHOP ROADMAP

- 01 Statistics & Data — population, sample, data types
- 02 Descriptive Statistics — centre, position and variability
- 03 Probability — events, counting and conditional probability
- 04 Probability Distributions — random variables, binomial, Poisson and normal
- 05 Sampling & Inference — sampling distributions, CLT, confidence intervals and hypothesis tests

Workshop pattern: concept → formula → numerical → interpretation

WHY STATISTICS MATTERS IN BUSINESS

- Statistics is the art and science of collecting, analysing, presenting and interpreting data.
- Accounting: analyse financial records
- Finance: measure risk and returns
- Marketing: understand customers
- Economics: study markets and trends

Analytics transforms data into insights for better decisions.

POPULATION, SAMPLE AND CENSUS

- Population: the entire set of elements of interest in a study. Example: all students in a college.
- Sample: a subset selected from the population. Example: 100 students selected for a survey.
- Census: collecting data from every unit of the population.
- Limitations of census: time-consuming, expensive, labour-intensive and unsuitable for destructive testing.

MEAN

- Mean is the sum of all the observations divided by the number of observations.

Mean: $\bar{x} = \frac{\sum x_i}{n}$ $x_i \rightarrow$ **lth observation in a data set of size n**

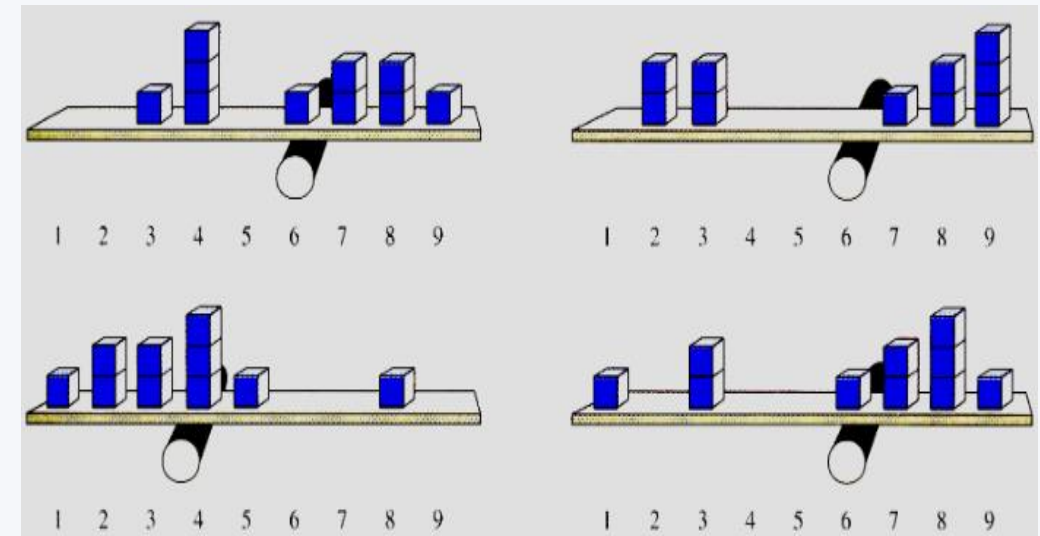
- Sample of 5 Class sizes = 46, 54, 42, 46, 32

$$\sum x_i = 220$$

$$n = 5$$

$$\bar{x} = 220 / 5 = 44 \text{ students}$$

Interpretation: average class size is 44 students.



MEDIAN

- Median is the value in the middle of a data set, when the data is arranged in ascending order.
- If ($n \rightarrow$ odd), the median is the middle value.

$$\text{Median} = \left(\frac{n+1}{2} \right) \text{ observation}$$

- If ($n \rightarrow$ even), the median is the average of the two middle values.

$$\text{Median} = \left(\frac{(n/2)(n/2 + 1)}{2} \right) \text{ observation}$$

Examples

- 52, 76, 100, 136, 186

$$n = 5$$

Therefore, the median hotel rate is

$$(5+1)/2 = 3^{\text{rd}} \text{ observation, i.e., } \mathbf{100}$$

- 2, 2, 3, 5, 6, 7, 7, 7, 8, 9.

$$n = 10$$

$$n/2 = 5^{\text{th}} \text{ value, i.e., } \mathbf{6}$$

$$(n/2 + 1) = 6^{\text{th}} \text{ value, i.e., } \mathbf{7}$$

Average of the middle two observations.

$$(6+7)/2 = 6.5$$

Therefore, the median is **6.5**

MODE

- Value of a data set that occurs with the greatest frequency.
- Example: 52, 76, 100, 136, 186, 196, 205, 150, 257, 264, 264, 280, 282, 283, 303, 313, 317, 317, 325, 373, 384, 384, 400, 402, 417, 422, 472, 480, 643, 693, 732, 749, 750, 791, 891.

Therefore, Mode(s) for hotel rates: 264, 317, 384

PERCENTILE

- Provides information about how the data is spread over the interval from the smallest to the largest value.
- $L_p = \left(\frac{P(n+1)}{100} \right)$

Example: 7 business graduate salaries: 5710,5755,5850,5880,5880,5890,5920.

Location of the 80th percentile.

$$L_{80} = \left(\frac{P(n+1)}{100} \right) = \left(\frac{80(7+1)}{100} \right) = \left(\frac{80 * 8}{100} \right) = 6.4$$

80th Percentile is 40% above the value in position 6, i.e., between the 6th and 7th value.

$$\begin{aligned} \text{Therefore, 80}^{\text{th}} \text{ Percentile} &= 5890 + 0.4(5920-5890) \\ &= 5890 + 0.4(30) \\ &= 5902 \end{aligned}$$

QUARTILE

- Divides the data set into 4 equal parts.
- Each part contains 25% of the observations.

Q1 First Quartile (25th Percentile)

Q2 Second Quartile (50th Percentile)

Q3 Third Quartile (75th Percentile)

- Example: 7 business graduate salaries: 5710,5755,5850,5880,5880,5890,5920.

Location of the 75th percentile.

$$L_{75} = \left(\frac{P(n+1)}{100} \right) = \left(\frac{75(7+1)}{100} \right) = \left(\frac{75 * 8}{100} \right) = 6^{\text{th}} \text{ observation}$$

75th Percentile is 5890.

MEASURES OF VARIABILITY

- Range: The difference between the smallest and largest observations.

$$\text{Range} = \text{Maximum Value} - \text{Minimum Value}$$

- Inter Quartile Range:

$$\text{IQR} = Q_3 - Q_1$$

- Example: No. of students in different sections of a school:

32,42,46,46,54.

$$Q_3 = \frac{75 (5+1)}{100} = \frac{75 * 6}{100} = 4.5$$

Therefore, $Q_3 = 46 + 0.5(54-46) = 50$

$$Q_1 = \frac{25 (5+1)}{100} = \frac{25 * 6}{100} = 1.5$$

Therefore, $Q_1 = 32 + 0.5(42-32) = 37$

$$\text{IQR} = Q_3 - Q_1 = 50 - 37 = 13$$

VARIANCE

- Difference between the value of each observation (X_i) and the mean ($\bar{X}$).

$$\sigma^2 = \frac{\sum (x - \mu)^2}{N} \quad \text{Population Variance}$$

$$s^2 = \frac{\sum (x - \bar{x})^2}{n - 1} \quad \text{Sample Variance}$$

- Example: No. of students in a class: 46,54,42,46,32

$$\text{Mean} = \frac{32 + 42 + 46 + 46 + 54}{5} = 44$$

$$\text{Variance } (s^2) = \frac{(32-44) + (42-44) + (46-44) + (46-44) + (54-44)}{5-1} = 64$$

STANDARD DEVIATION

- Positive square root of variance.
- Measures the average distance of a data point from the mean.

$$s = \sqrt{\frac{\sum (x - \bar{x})^2}{n - 1}}$$

- Example: No. of students in a class: 46,54,42,46,32

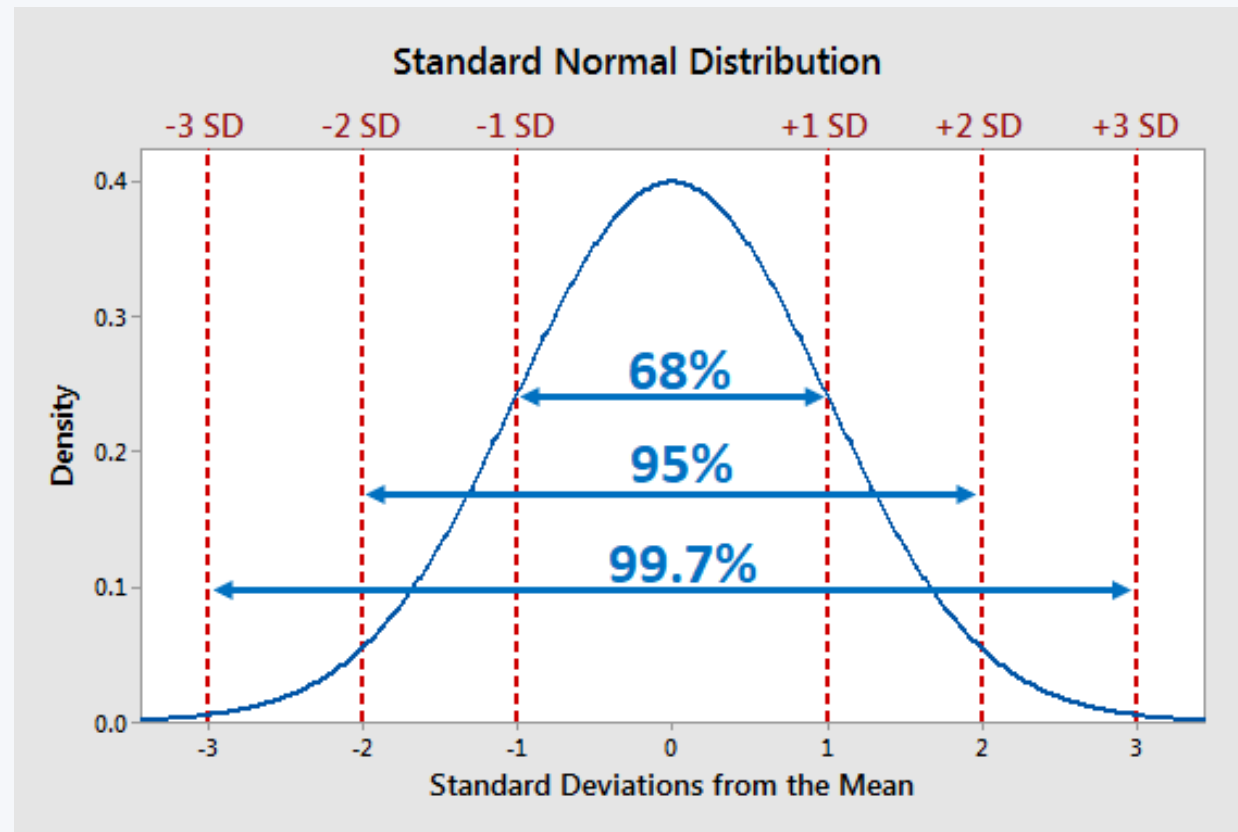
$$\text{Mean} = \frac{32 + 42 + 46 + 46 + 54}{5} = 44$$

$$\text{Variance } (s^2) = \frac{(32-44)^2 + (42-44)^2 + (46-44)^2 + (46-44)^2 + (54-44)^2}{5-1} = 64$$

$$\text{St. Dev. } (s) = 8$$

EMPIRICAL RULE

- When the data exhibits a symmetric bell-shaped distribution, the empirical rule can be used to determine the percentage of data values that must be within a specific number of St. Dev. Of the mean



Probability

Definition: Probability is a numerical measure (between 0 and 1) of the likelihood that a particular event will occur.

Three Methods of Assigning Probability

1 Classical *(A Priori)*

Based on equally likely outcomes, known in advance.

$P(E) = \text{Favorable outcomes} / \text{Total outcomes}$

Used when outcomes are symmetric, e.g. tossing a fair coin or rolling a die.

2 Relative Frequency *(Empirical)*

Based on observed data from repeated trials or historical records.

$P(E) = \text{Number of times E occurs} / \text{Total trials}$

Used when outcomes are not equally likely, e.g. default rates from past loan data.

3 Subjective *(Personal Judgment)*

Based on personal belief, experience, or expert judgment, not on formal computation.

Used when data is limited or unique events occur, e.g. estimating the probability a new product succeeds.

PROBABILITY – KEY FORMULAS

ADDITION RULE

General: $P(A \text{ or } B) = P(A) + P(B) - P(A \cap B)$

Mutually exclusive: $P(A \text{ or } B) = P(A) + P(B)$,
since $P(A \cap B) = 0$

CONDITIONAL PROBABILITY

A given B: $P(A | B) = P(A \cap B) / P(B)$

B given A: $P(B | A) = P(A \cap B) / P(A)$

LAW OF CONDITIONAL PROBABILITY

Formula: $P(X | Y) = P(X \cap Y) / P(Y)$

Equivalent: $P(X | Y) = P(X) \cdot P(Y | X) / P(Y)$

COMPLEMENT RULE

The complement of an event A is the event that A does not happen. It is written A' , A^c or $\bar{A}$.
 $P(A') = 1 - P(A)$

INDEPENDENT EVENTS

Condition: $P(A | B) = P(A)$

Equivalent: $P(B | A) = P(B)$

SPECIAL MULTIPLICATION LAW

If independent: $P(A \cap B) = P(A) \cdot P(B)$

Meaning: Occurrence of one does not affect the other

EVENT TYPES

Simple event: One characteristic

Joint event: Two or more characteristics

Mutually exclusive: Cannot occur simultaneously

Collectively exhaustive: Events cover the entire sample space

KEY NOTATION

$P(A)$: Probability of A

$P(A \text{ and } B)$: Joint probability

$P(A | B)$: Probability of A given B

$A \cap B$: A and B

$A \cup B$: A or B

QUICK REMINDERS

Mutually exclusive: $P(A \cap B) = 0$

Independent: $P(A | B) = P(A)$

Independent multiplication: $P(A \cap B) = P(A)P(B)$

Conditional: Always divide by the given event's probability

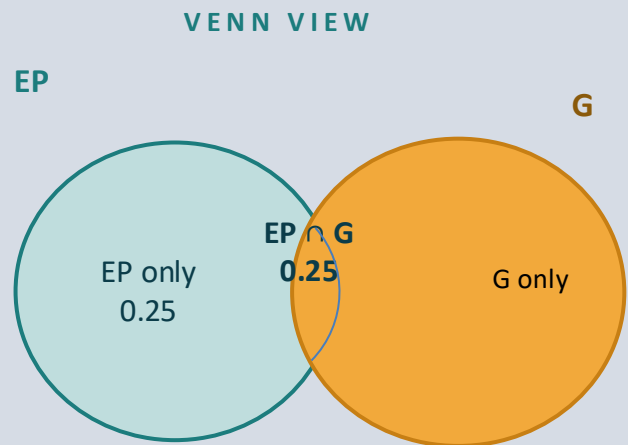
Question 1 · At least one of two events

QUESTION

It was estimated that 50% of all juniors in a campus were seriously concerned about employment prospects, 25% were seriously concerned about grades, and 25% were seriously concerned about both. What is the probability that a randomly chosen junior from this campus is seriously concerned about at least one of these two events?

Question 1 · At least one of two events

Solution:



GIVEN

$$P(\text{EP}) = 0.50$$

$$P(\text{G}) = 0.25$$

$$P(\text{EP} \cap \text{G}) = 0.25$$

SOLUTION

$$P(\text{EP} \cup \text{G}) = P(\text{EP}) + P(\text{G}) - P(\text{EP} \cap \text{G})$$

$$= 0.50 + 0.25 - 0.25$$

$$= 0.5$$

ANSWER

0.5

50%

Question 2 • Subscription campaign

QUESTION

The publisher of a popular financial magazine has decided to undertake a campaign in an effort to attract new subscribers. Market research analysts in this company believe that there is a 1 in 4 chance that the increase in the number of new subscriptions resulting from this campaign will be less than 3000, and there is a 1 in 3 chance that the increase in the number will be between 3000 and 5000. What is the probability that the increase in the number of new subscriptions resulting from this campaign will be less than 3000 or more than 5000?

Question 2 · Subscription campaign

Solution:

Less than 3,000

$$\frac{1}{4}$$

Between 3,000 and 5,000

$$\frac{1}{3}$$

More than 5,000

$$\frac{5}{12}$$

$$= 1 - \left(\frac{1}{4} + \frac{1}{3}\right)$$

The two ranges cannot occur together, so $P(A \cap B) = 0$ and the probabilities simply add.

$$P(<3,000 \text{ or } >5,000) = \frac{1}{4} + \frac{5}{12} = \frac{3}{12} + \frac{5}{12} = \frac{8}{12}$$

ANSWER

$$\frac{2}{3} \approx 0.667$$

Question 3 · MBA and non-MBA interviews

QUESTION

An MBA AND NON-MBA graduate are appearing for an interview for two vacancies in the same post. The probability of MBA graduate is being selected is $\frac{1}{7}$ and that of non-MBA is $\frac{1}{5}$. Find the probability that a) both b) only one c) none d) at least one will be selected.

Question 3 · MBA and non-MBA interviews

QUESTION

An MBA AND NON-MBA graduate are appearing for an interview for two vacancies in the same post. The probability of MBA graduate is being selected is $1/7$ and that of non-MBA is $1/5$. Find the probability that a) both b) only one c) none d) at least one will be selected.

Given: $P(\text{MBA}) = 1/7$, $P(\text{Non-MBA}) = 1/5 \rightarrow P(\text{MBA}') = 6/7$, $P(\text{Non-MBA}') = 4/5$

a Both

$$1/7 \times 1/5 = 1/35$$

$$1/35 \approx 0.0286$$

b Only one

$$(1/7 \times 4/5) + (6/7 \times 1/5) \\ = 4/35 + 6/35 = 10/35$$

$$2/7 \approx 0.2857$$

c None

$$6/7 \times 4/5 = 24/35$$

$$24/35 \approx 0.6857$$

d At least one

$$1/7 + 1/5 - 1/35 \\ = (5 + 7 - 1)/35$$

$$11/35 \approx 0.3143$$

Check: $P(\text{both}) + P(\text{only one}) + P(\text{none}) = 1/35 + 10/35 + 24/35 = 1$

Question 4 · CONDITIONAL PROBABILITY

QUESTION

Two cards are drawn from a 52-card deck without replacement. Given that the first card is an Ace, what is the probability that the second card is also an Ace?

A = second card is an Ace

B = first card is an Ace

Solution :

Given:

$$P(B) = 4/52 = 1/13$$

$$P(A \cap B) = (4/52) \times (3/51) = 1/221$$

$$P(A | B) = (1/221) / (1/13) = 13/221$$

$$= 1/17$$

Example: CONFIRM bids for two jobs

Problem

CONFIRM, a consulting firm, is bidding for two jobs, one with each of two large multinational corporations. The probability of getting the job with firm A is 0.45. If it gets the job with A, there is a 0.90 probability that firm B will also give it the consulting job. What are CONFIRM's chances of getting both jobs?

A = gets job with firm A

B = gets job with firm B

Solution

Given: $P(A) = 0.45$ and $P(B | A) = 0.90$

From the law: $P(A \cap B) = P(A) \times P(B | A)$

$$P(A \cap B) = 0.45 \times 0.90 = 0.405$$

CONFIRM has a 40.5% chance of getting both jobs.

Question 5· The ideal institution

QUESTION

A graduate is looking for a management institution which has following characteristics; good infrastructure-the probability of getting admission in such institution is one in fifty, good placement record-probability of getting admission in such institution is one in hundred and good faculty support-probability of getting admission in such institution is one in thousand. Find the probability of graduate is getting admitted in such institution which possesses all the three characteristics. (assuming all the three are independent).

Good infrastructure

1/50

×

Good placement record

1/100

×

Good faculty support

1/1000

=

ANSWER

1/5,000,000

= 0.0000002 (2×10^{-7})

Independent events: $P(A \cap B \cap C) = P(A) \times P(B) \times P(C)$

Definition and probability density function

CONTINUOUS RANDOM VARIABLE

Takes any value within a range. $E(X) = \int x f(x) dx$ $V(X) = \int (x - \mu)^2 f(x) dx$

A continuous random variable X follows a normal distribution if it has the probability density function (PDF):

$$f(x) = 1 / (\sigma\sqrt{2\pi}) \cdot e^{-\frac{1}{2} ((x - \mu)/\sigma)^2}$$

for $-\infty < x < \infty$



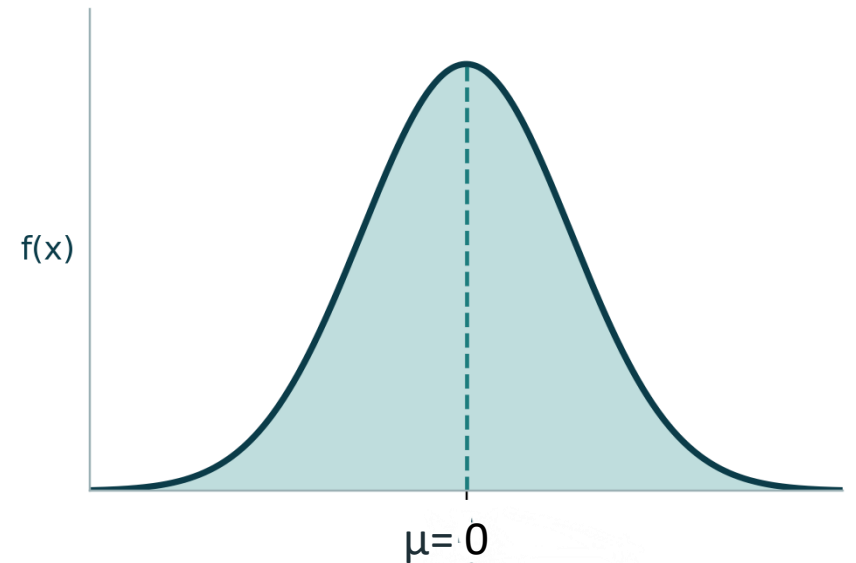
Mean $(-\infty < \mu < \infty)$



Standard deviation $(\sigma > 0)$



Range $-\infty < x < \infty$



Plotting x on the x -axis against $f(x)$ on the y -axis gives the bell-shaped normal curve.

Properties of the normal curve

Total area = 1

1

The total area under the normal curve is always unity.

The mean splits the curve in half

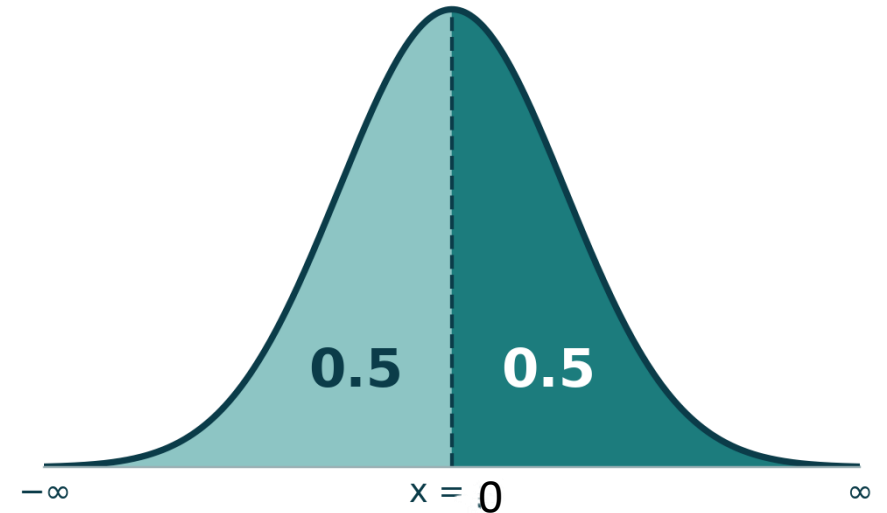
2

The mean (μ) is the point that divides the curve into two equal parts, 0.5 on each side.

Mean = Median = Mode

3

All three are always equal, and therefore the skewness is zero.



Each half of the curve holds an area of 0.5.

The standard normal distribution

If $X \sim N(\mu, \sigma^2)$ then

$$Z = (X - \mu) / \sigma$$

follows the standard normal distribution, $Z \sim N(0, 1)$.

MEAN

0

VARIANCE

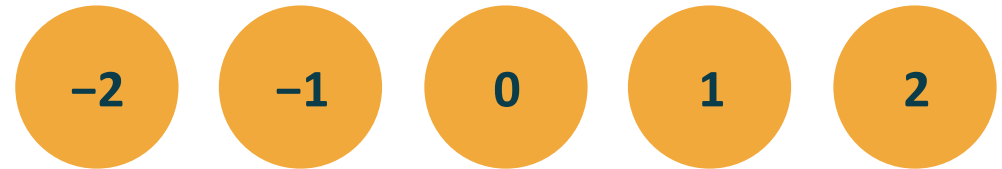
1

Worked conversion for $X \sim N(30, 25), \sigma = 5$

X (minutes)

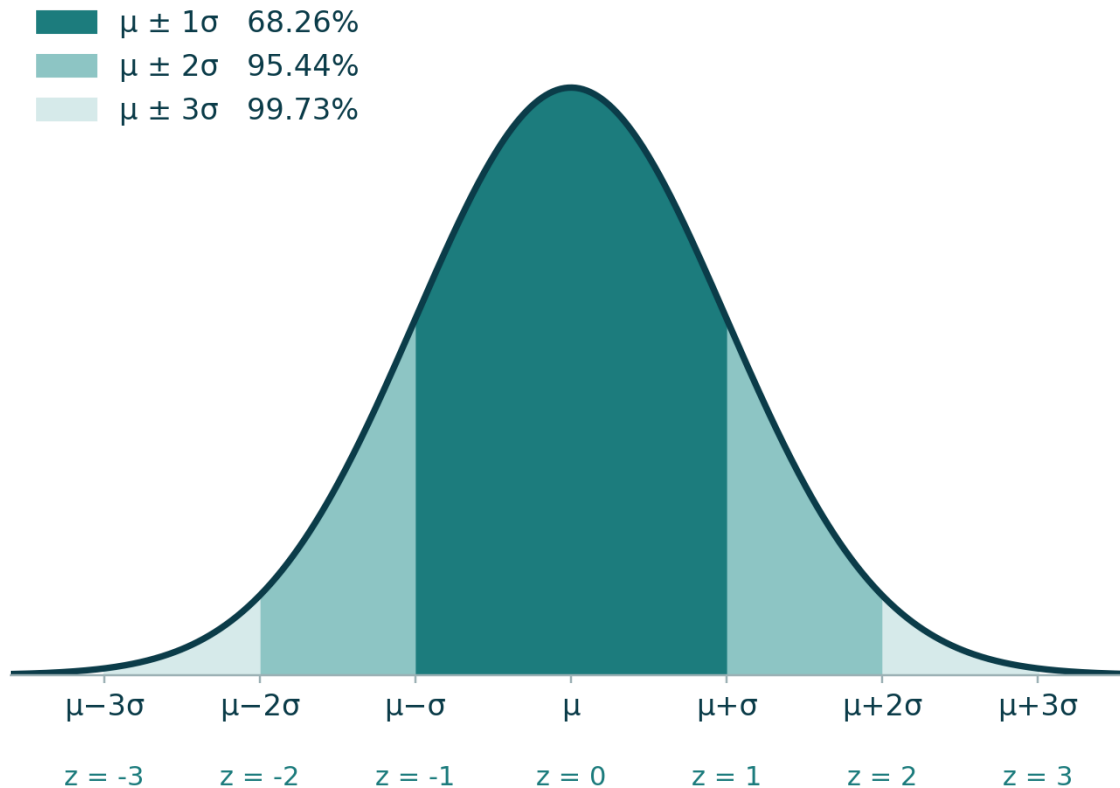


$z = (X - 30) / 5$



Converting to z lets a single table serve every normal distribution.

Area within 1, 2 and 3 standard deviations



68.26%

$\mu \pm 1\sigma$

$$P(\mu - \sigma < X < \mu + \sigma)$$

$$= P(-1 < Z < 1) = 0.6826$$

95.44%

$\mu \pm 2\sigma$

$$P(\mu - 2\sigma < X < \mu + 2\sigma)$$

$$= P(-2 < Z < 2) = 0.9544$$

99.73%

$\mu \pm 3\sigma$

$$P(\mu - 3\sigma < X < \mu + 3\sigma)$$

$$= P(-3 < Z < 3) = 0.9973$$

Case-solving time of a business school student

QUESTION

An average student in the leading business school takes on an average 30 minutes to solve the case given by operations faculty with a variance of 25 minutes. Assuming that the time taken to solve the case follows normal distribution, if a student is taken at random find the probability that he takes

- a Less than or equal to 30 minutes
- b Less than or equal to 40 minutes
- c Between 40 and 45 minutes
- d Between 25 and 35 minutes
- e Between 20 and 25 minutes
- f More than 35 minutes

GIVEN

Mean

$$\mu = 30 \text{ min}$$

Variance

$$\sigma^2 = 25, \sigma = 5$$

Distribution

$$X \sim N(30, 25)$$

Standardise

$$Z = (X - 30) / 5$$

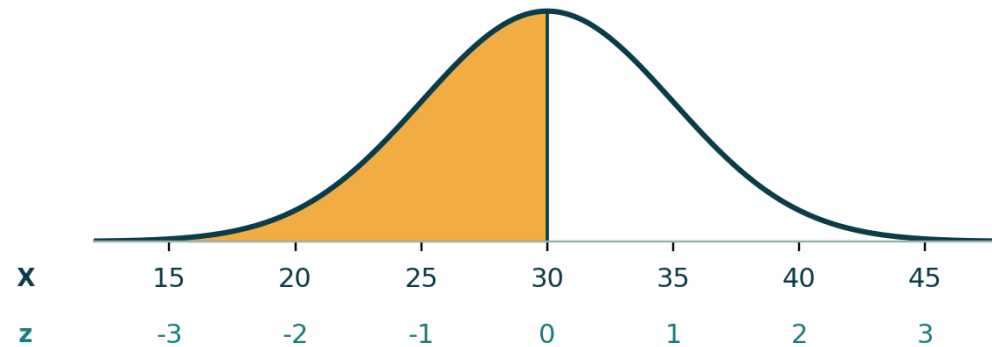
Solution · parts (a) and (b)

a $P(X \leq 30)$

$$z = (30 - 30) / 5 = 0$$

$$P(X \leq 30) = P(Z \leq 0)$$

Half of the area lies to the left of the mean.



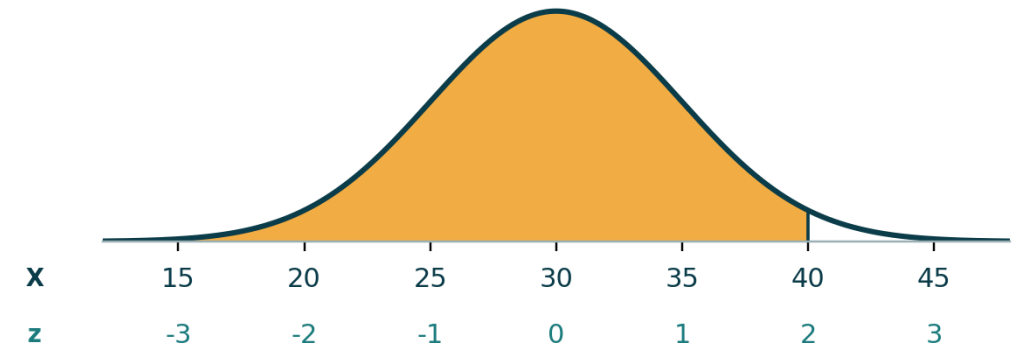
Answer: 0.5

b $P(X \leq 40)$

$$z = (40 - 30) / 5 = 2$$

$$P(X \leq 40) = P(Z \leq 2) = 0.5 + P(0 < Z < 2)$$

$$= 0.5 + 0.4772$$



Answer: 0.9772

Solution · parts (c) and (d)

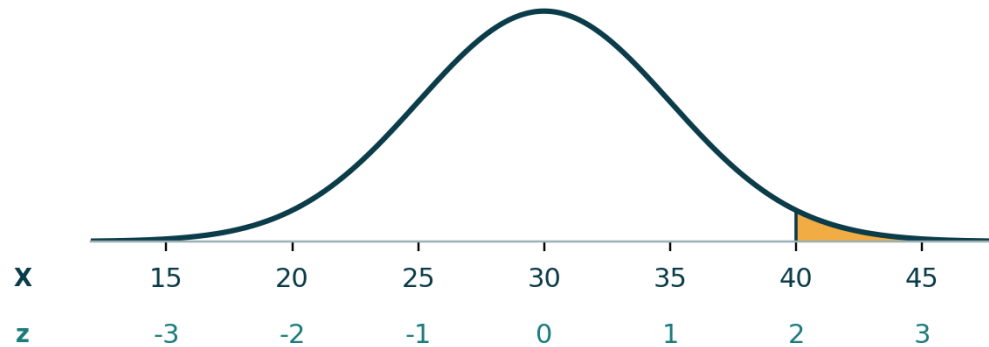
c $P(40 < X < 45)$

$$x = 40 \rightarrow z = (40 - 30)/5 = 2$$

$$x = 45 \rightarrow z = (45 - 30)/5 = 3$$

$$P(2 < Z < 3) = P(0 < Z < 3) - P(0 < Z < 2)$$

$$= 0.4987 - 0.4772$$



Answer: 0.0215

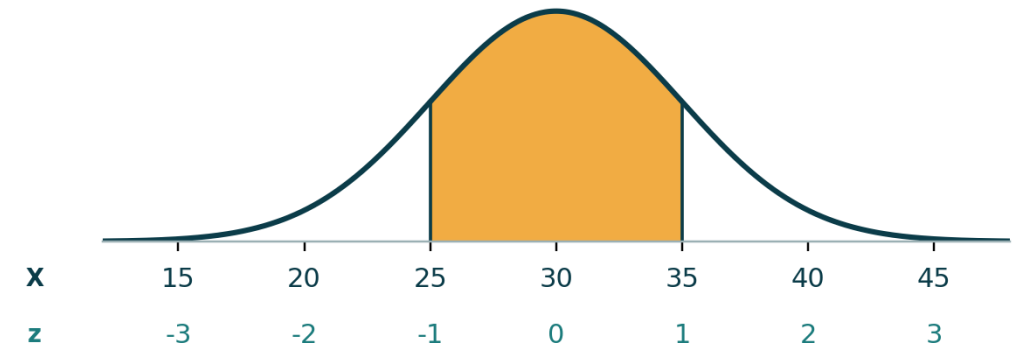
d $P(25 < X < 35)$

$$x = 25 \rightarrow z = (25 - 30)/5 = -1$$

$$x = 35 \rightarrow z = (35 - 30)/5 = 1$$

$$P(-1 < Z < 1) = 2 \times P(0 < Z < 1)$$

$$= 2 \times 0.3413$$



Answer: 0.6826

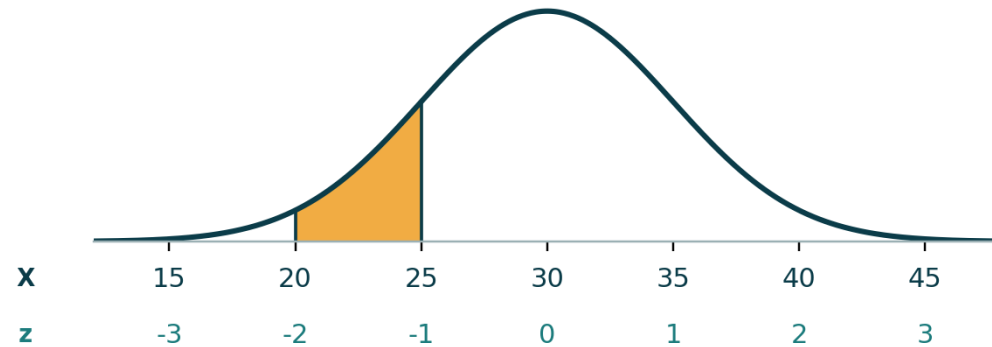
Solution · parts (e) and (f)

e $P(20 < X < 25)$

$$x = 20 \rightarrow z = (20 - 30)/5 = -2$$

$$x = 25 \rightarrow z = (25 - 30)/5 = -1$$

$$\begin{aligned} P(-2 < Z < -1) &= P(0 < Z < 2) - P(0 < Z < 1) \\ &= 0.4772 - 0.3413 \end{aligned}$$

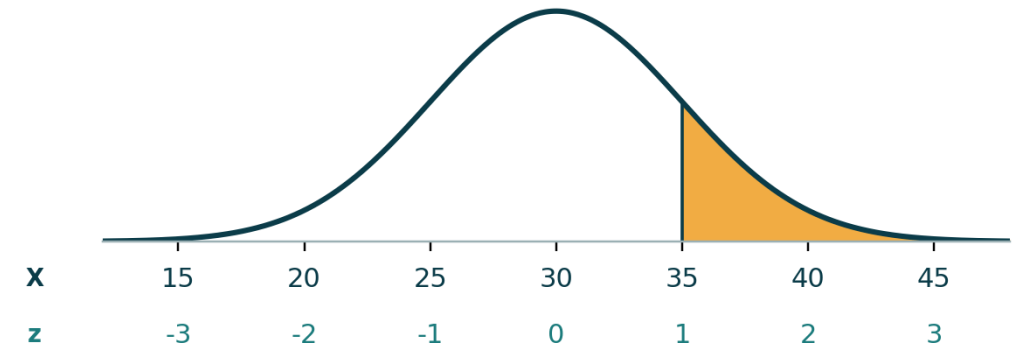


Answer: 0.1359

f $P(X > 35)$

$$z = (35 - 30) / 5 = 1$$

$$\begin{aligned} P(X > 35) &= P(Z > 1) = 0.5 - P(0 < Z < 1) \\ &= 0.5 - 0.3413 \end{aligned}$$



Answer: 0.1587

Standard normal table · z = 0.0 to 1.7

z	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
0.0	0.0000	0.0040	0.0080	0.0120	0.0160	0.0190	0.0239	0.0279	0.0319	0.0359
0.1	0.0398	0.0438	0.0478	0.0517	0.0557	0.0596	0.0636	0.0675	0.0714	0.0753
0.2	0.0793	0.0832	0.0871	0.0910	0.0948	0.0987	0.1026	0.1064	0.1103	0.1141
0.3	0.1179	0.1217	0.1255	0.1293	0.1331	0.1368	0.1406	0.1443	0.1480	0.1517
0.4	0.1554	0.1591	0.1628	0.1664	0.1700	0.1736	0.1772	0.1808	0.1844	0.1879
0.5	0.1915	0.1950	0.1985	0.2019	0.2054	0.2088	0.2123	0.2157	0.2190	0.2224
0.6	0.2257	0.2291	0.2324	0.2357	0.2389	0.2422	0.2454	0.2486	0.2517	0.2549
0.7	0.2580	0.2611	0.2642	0.2673	0.2704	0.2734	0.2764	0.2794	0.2823	0.2852
0.8	0.2881	0.2910	0.2939	0.2969	0.2995	0.3023	0.3051	0.3078	0.3106	0.3133
0.9	0.3159	0.3186	0.3212	0.3238	0.3264	0.3289	0.3315	0.3340	0.3365	0.3389
1.0	0.3413	0.3438	0.3461	0.3485	0.3508	0.3513	0.3554	0.3577	0.3529	0.3621
1.1	0.3643	0.3665	0.3686	0.3708	0.3729	0.3749	0.3770	0.3790	0.3810	0.3830
1.2	0.3849	0.3869	0.3888	0.3907	0.3925	0.3944	0.3962	0.3980	0.3997	0.4015
1.3	0.4032	0.4049	0.4066	0.4082	0.4099	0.4115	0.4131	0.4147	0.4162	0.4177
1.4	0.4192	0.4207	0.4222	0.4236	0.4251	0.4265	0.4279	0.4292	0.4306	0.4319
1.5	0.4332	0.4345	0.4357	0.4370	0.4382	0.4394	0.4406	0.4418	0.4429	0.4441
1.6	0.4452	0.4463	0.4474	0.4484	0.4495	0.4505	0.4515	0.4525	0.4535	0.4545
1.7	0.4554	0.4564	0.4573	0.4582	0.4591	0.4599	0.4608	0.4616	0.4625	0.4633

HOW TO READ IT

Rows give z to one decimal, columns give the second decimal. Each value is the area between 0 and z, $P(0 < Z < z)$.

z = 1.25 → row 1.2, column 0.05 → 0.3944

PRINTED VALUES

Highlighted cells are copied as printed on the sheet, but differ from the standard table: 0.0190 (0.0199), 0.2969 (0.2967), 0.3513 (0.3531), 0.3529 (0.3599).

Standard normal table · z = 1.8 to 3.4

z	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
1.8	0.4641	0.4649	0.4656	0.4664	0.4671	0.4678	0.4686	0.4693	0.4699	0.4706
1.9	0.4713	0.4719	0.4726	0.4732	0.4738	0.4744	0.4750	0.4756	0.4761	0.4767
2.0	0.4772	0.4778	0.4783	0.4788	0.4793	0.4798	0.4803	0.4808	0.4812	0.4817
2.1	0.4821	0.4826	0.4830	0.4834	0.4838	0.4842	0.4846	0.4850	0.4854	0.4857
2.2	0.4861	0.4864	0.4868	0.4871	0.4875	0.4878	0.4881	0.4884	0.4887	0.4890
2.3	0.4893	0.4896	0.4898	0.4901	0.4904	0.4906	0.4909	0.4911	0.4913	0.4916
2.4	0.4918	0.4920	0.4922	0.4925	0.4927	0.4929	0.4931	0.4932	0.4934	0.4936
2.5	0.4938	0.4940	0.4941	0.4943	0.4945	0.4946	0.4948	0.4949	0.4951	0.4952
2.6	0.4953	0.4955	0.4956	0.4957	0.4959	0.4960	0.4961	0.4962	0.4963	0.4964
2.7	0.4965	0.4966	0.4967	0.4968	0.4969	0.4970	0.4971	0.4972	0.4973	0.4974
2.8	0.4974	0.4975	0.4976	0.4977	0.4977	0.4978	0.4979	0.4979	0.4980	0.4981
2.9	0.4981	0.4982	0.4982	0.4983	0.4984	0.4984	0.4985	0.4985	0.4986	0.4986
3.0	0.4987	0.4987	0.4987	0.4988	0.4988	0.4989	0.4989	0.4989	0.4990	0.4990
3.1	0.4990	0.4991	0.4991	0.4991	0.4992	0.4992	0.4992	0.4992	0.4993	0.4993
3.2	0.4993	0.4993	0.4994	0.4994	0.4994	0.4994	0.4994	0.4995	0.4995	0.4995
3.3	0.4995	0.4995	0.4995	0.4996	0.4996	0.4996	0.4996	0.4996	0.4996	0.4997
3.4	0.4997	0.4997	0.4997	0.4997	0.4997	0.4997	0.4997	0.4997	0.4997	0.4998

VALUES USED IN QUESTION 2

z = 1.00
0.3413

z = 2.00
0.4772

z = 3.00
0.4987

Beyond z = 3.4 the area is effectively 0.5.

Binomial Distribution

A binomial distribution describes the probability of obtaining a certain number of successes in a fixed number of independent trials.

Binomial Distribution - Properties

- Identical trials: There is a repeated, finite number of trials, say n times.
- Two outcomes: Each trial has only two possible outcomes: Success or Failure.
- Constant probability: The probability of success (p) and failure ($q=1-p$) remain constant.
- Independent trials: The outcome of one trial does not affect the outcome of another.

Binomial Distribution - Formula

Binomial Distribution Formula

$$P(x) = \binom{n}{x} p^x q^{n-x} = \frac{n!}{(n-x)!x!} p^x q^{n-x}$$

where

n = the number of trials (or the number being sampled)

x = the number of successes desired

p = probability of getting a success in one trial

$q = 1 - p$ = the probability of getting a failure in one trial

Mean: $E(x) = np$

Variance: $V(x) = npq$

Binomial Distribution

Ques 1. Suppose a salesman is successful in obtaining an order in one out of every 5 calls he makes.

In a normal working day he contacts 10 firms. What is the probability that:

- (a) He gets 3 orders
- (b) He gets no orders
- (c) He gets at least 2 orders.

$$n = 10 \text{ [no. of calls]}$$

$$p = 1/5 = 0.2 \text{ (probability of success)}$$

$$q = 1 - p = 1 - 0.2 = 0.8 \text{ (probability of failure)}$$

(a) Exactly 3 orders

$$P(X=3) = {}^{10}C_3 \cdot (0.2)^3 \cdot (0.8)^{(10-3)}$$
$$= 120 \times 0.008 \times 0.2097152$$

$$= 0.2037152 \text{ . Ans}$$

$$\underline{P(X=3) \approx 0.2013}$$

(b) No orders means $X=0$

$$P(X=0) = {}^{10}C_0 \cdot (0.2)^0 \cdot (0.8)^{(10-0)}$$

$$= 1 \cdot (1) \cdot (0.10737)$$

$$P(X=0) = 0.10737$$

$$\underline{P(X=0) \approx 0.1074 \text{ . Ans}}$$

$$\textcircled{c} \quad p(x \geq 2) = 1 - p(x < 2)$$

$$= 1 - [p(x=0) + p(x=1)]$$

$$= 1 - [0.1074 + {}^{10}C_1 (0.2)^1 (0.8)^{10-1}]$$

$$= 1 - [0.1074 + 10 \times 0.2 \times 0.1342]$$

$$= 1 - [0.1074 + 0.2684]$$

$$\underline{p(x \geq 2) = 0.6242} \quad \text{Ans.}$$

Binomial Distribution

Ques 2. If the mean and variance of the given Binomial distribution are 3 and 2, find $P(X \geq 2)$.

$$\text{Mean, } \mu = n \cdot p = 3$$

$$\text{Variance, } \sigma^2 = n \cdot p \cdot q = 2$$

$$\frac{n p q}{n p} = \frac{2}{3}$$

$$\underline{q = 2/3}$$

$$p + q = 1$$

$$\underline{p = 1 - 2/3 = 1/3}$$

$$n \cdot p = 3$$

$$n \cdot 1/3 = 3$$

$$\underline{n = 9}$$

$$P(X \geq 2) = 1 - [P(X=0) + P(X=1)]$$

$$P(X=0) = {}^9C_0 \cdot (1/3)^0 \cdot (2/3)^{9-0} = 0.026012$$

$$P(X=1) = {}^9C_1 (1/3)^1 \cdot (2/3)^{9-1} = 0.117052$$

$$P(X \geq 2) = [1 - (0.026012) + (0.117052)] = 0.856936$$

$$\underline{P(X \geq 2) \approx 0.8569} \text{ . Ans .}$$

Poisson Distribution

A Poisson distribution is used for events that occur at a constant average rate over a given interval of time or space.

Poisson Distribution - Properties

- Identical trials: There are a repeated, infinite number of trials.
- Two outcomes: Each trial has two outcomes: Success or Failure.
- Very low probability: The probability of success (p) is very low.
- Independent trials: All trials are independent.

Poisson Distribution - Formula

$$P(X = x) = \frac{\lambda^x e^{-\lambda}}{x!}$$

where

$x = 0, 1, 2, 3, \dots$

λ = mean number of occurrences in the interval

e = Euler's constant ≈ 2.71828

Mean: $E(x) = m = np$

Variance: $V(x) = m = np$

Poisson Distribution

Ques 1. A building society branch manager notices that over a long period of time the number of people using an automated cash point on a Saturday morning is on average 30 people per hour. What is the probability that in, say, a 10 minutes period:

(a) none

(b) three people use the machine?

The given rate is 30 people per 60 minutes, for 10 minutes:

$$\lambda = 30 \times \left(\frac{10}{60}\right) = 5$$

(a) None:

$$X=0$$

$$\lambda = 5$$

$$P(X=0) = \frac{(e)^{-5} \cdot (5)^0}{0!} \quad [0! = 1]$$

$$P(X=0) = 0.0067379$$

$$P(X=0) \approx 0.00674 \text{ .Ans.}$$

(b) Exactly 3

$$X=3$$

$$\lambda = 5$$

$$P(X=3) = \frac{(e)^{-5} \cdot (5)^3}{3!}$$

$$P(X=3) = 0.1403739$$

$$P(X=3) \approx 0.1404 \text{ .Ans.}$$

Poisson Distribution

Ques 2. In a study of business activity at a restaurant it is found that the average number of customers arriving between 10 and 11 pm on a particular day is 5. Using a Poisson distribution answer the following:

- (a) What is the probability that only one customer arrives?
- (b) at least two will arrive.

(a) Exactly one customer

$$X = 1$$

$$\lambda = 5$$

$$P(X=1) = \frac{(e)^{-5} \cdot (5)^1}{1!}$$

$$P(X=1) = 0.336897$$

$$P(X=1) \approx 0.3369 \quad \text{Ans.}$$

(b) ~~Ex~~ At least two customers

$$P(X \geq 2) = 1 - [P(X=0) + P(X=1)]$$

$$P(X=0) = \frac{(e)^{-5} \cdot (5)^0}{0!} = 0.0067379$$

$$P(X=1) = 0.336897$$

$$P(X \geq 2) = 1 - [0.0067379 + 0.336897]$$

$$P(X \geq 2) = 0.9595723$$

$$P(X \geq 2) \approx 0.9596 \quad \text{Ans.}$$

Binomial vs Poisson

Feature	Binomial	Poisson
Number of trials	Fixed (n)	Infinite
Probability	Constant (p)	Very low (p)
Mean	np	np or m
Variance	npq	np or m
Distribution shape	Binary outcomes	Rare events

INFERENCEAL STATISTICS

Hypothesis Testing — A practical 5-step method

Why do we do hypothesis testing?

To use sample evidence to decide whether there is enough evidence to question a claim about a population.

The 5 steps for a hypothesis testing problem

Use the same flow every time.

- 1 Formulate H_0 and H_1** State the null and alternative hypotheses. Then identify one-tail or two-tail.
- 2 Find the level of significance** Usually $\alpha = 0.05$ unless the question gives another value.
- 3 Find the test statistic** Use the appropriate z or t formula from the information given.
- 4 Find the critical value** Look up z_{α} / $z_{\alpha/2}$ or the appropriate t critical value in the table.
- 5 Conclude** Compare the calculated statistic with the critical value and decide whether to reject H_0 .

Step 1 — Formulate H_0 and H_1

The alternative hypothesis tells you which tail to use.

How to spot the tail

ONE-TAILED

- H_1 uses $>$ or $<$.
- Words like “higher”, “greater”, “increased”, “less”, “lower”, or “decreased” point in one direction.
- All of α is in one tail.

Example: $H_1: \mu > 500$

TWO-TAILED

- H_1 uses $\neq$.
- Words like “different”, “not equal”, or “changed” allow either direction.
- α is split into two tails: $\alpha/2$ in each.

Example: $H_1: \mu \neq 500$

Step 1 — Quick note: z-test or t-test?

Choose the test from the information the question gives you.

Use a Z-test

- The population standard deviation σ is known/given.
- For a one-sample mean test, the standard error is $\sigma/\sqrt{n}$.

Use a t-test

- The population standard deviation σ is unknown.
- The question gives a sample standard deviation s instead.
- Use the relevant t distribution and degrees of freedom.

Easy check: σ given $\rightarrow$ think Z. s given instead of $\sigma \rightarrow$ think t.

Step 2 & 3 — Significance level and test statistic

2. Find the level of significance

α is the chosen significance level. Commonly $\alpha = 0.05$ or 0.01 .

3. Find the test statistic

$$z = (\bar{x} - \mu) / (\sigma / \sqrt{n})$$

$\bar{x}$ = sample mean • μ = population mean under H_0 • σ = population standard deviation • n = sample size

Important: n is the SAMPLE SIZE — not the sample mean.

Step 4 — Find the critical value

Look up the appropriate z or t table.

Standard normal (Z) critical values

α	One-tailed: z_α	Two-tailed: $z_{\alpha/2}$
0.05	1.645	± 1.96
0.01	2.326	± 2.576

For a t-test:

look up the t critical value using the significance level, one/two tails, and the appropriate degrees of freedom (df).

For two-tailed tests, split α into $\alpha/2$ in each tail.

Step 5 — Compare and conclude

The decision rule depends on whether the test is one-tailed or two-tailed.

ONE-TAILED

Right-tailed ($H_1: \mu > \mu_0$)

Reject H_0 if $z > z_\alpha$

Do not reject H_0 if $z \leq z_\alpha$

Left-tailed ($H_1: \mu < \mu_0$)

Reject H_0 if $z < -z_\alpha$

Do not reject H_0 if $z \geq -z_\alpha$

TWO-TAILED

$H_1: \mu \neq \mu_0$

Reject H_0 if $|z| > z_{\alpha/2}$

Do not reject H_0 if $|z| \leq z_{\alpha/2}$

Same logic for t-tests — compare the calculated t with the appropriate t critical value.

How to write the final conclusion

State the statistical decision, then connect it back to the claim.

If you REJECT H_0 “There is sufficient evidence at the α level to support the alternative claim.”

If you DO NOT REJECT H_0 “There is insufficient evidence at the α level to support the alternative claim.”

Claim link: End by saying what the sample evidence lets us infer about the population claim in the question.

Question 1 — Two-sample hypothesis test

Read the wording carefully before choosing the tail and test.

1. For a random sample of 125 British entrepreneurs the mean number of job changes was 1.91 and the sample standard deviation was 1.32. For an independent random sample of 86 British corporate managers the mean number of job changes was 0.21 and the sample standard deviation was 0.53. Test the null hypothesis that the population means are equal against the alternative that the mean number of job changes is higher for British entrepreneurs than for British corporate managers. Also, find 95% confidence interval for difference between means.

Hint: identify H_0/H_1 first, then decide whether this is one-tailed or two-tailed and whether z or t is appropriate.

H_0 / H_1

α

Test statistic

Critical value + conclusion

Critical Values from the t Distribution (continued)

Values of α for one-tailed test and $\alpha/2$ for two-tailed test

df	t.100	t.050	t.025	t.010	t.005	t.001
50	1.299	1.676	2.009	2.403	2.678	3.261
60	1.296	1.671	2.000	2.390	2.660	3.232
70	1.294	1.667	1.994	2.381	2.648	3.211
80	1.292	1.664	1.990	2.374	2.639	3.195
90	1.291	1.662	1.987	2.368	2.632	3.183
100	1.290	1.660	1.984	2.364	2.626	3.174
150	1.287	1.655	1.976	2.351	2.609	3.145
200	1.286	1.653	1.972	2.345	2.601	3.131
∞	1.282	1.645	1.960	2.326	2.576	3.090

Note: As $df \rightarrow \infty$ the t distribution approaches the standard normal (z) distribution. For large samples ($df \geq 30$) the z critical values are often used as approximations.
The values in the ∞ row are the familiar z critical values (1.645 for $\alpha=0.05$ one-tail, 1.960 for $\alpha=0.025$ one-tail / 0.05 two-tail, etc.).

Question 3 — One-sample hypothesis test

Read the wording carefully before choosing the tail and test.

3. Mr. X owns a retail store where he plans to give individual arriving customers a short verbal commercial message regarding the products for sale. Owner asks you to determine if this will boost the mean sales transaction level μ , currently at \$5. You may assume that the standard deviation per transaction is \$1. To monitor, a random sample of 200 customers were given this special treatment and determined that the mean sales was \$5.37. Formulate this as hypothesis and test whether mean sales have increased at 5% level. Also, find 95% confidence interval for the mean.

Hint: look for the direction of the claim and check whether σ or s is given.

H_0 / H_1 + tail

α

Test statistic

Critical value + conclusion + CI

Number 3

$\mu = 5$
 $\sigma = 1$
 $n = 200$
 $\bar{x} = 5.37$
 $\text{los} = 5\%$

Test of μ

- σ (known) (Z-test)
- σ (unknown) (t-test)

NH $H_0: \mu = \$5$
 HA $H_1: \mu > \$5$

ie the mean sales transaction per customer has increased from current level of \$5 due to short verbal commercial msg given to the customer.

$\text{los} = 5\%$ $\alpha = 5\%$

TS = $Z = \frac{\bar{x} - \mu}{\sigma / \sqrt{n}} \sim N(0,1)$

$= \frac{5.37 - 5}{1 / \sqrt{200}} = 5.23$

M	T	W	T	F	S	S
Page No.:						YOUVA
Date:						

Confidence Interval

$$\bar{X} \pm M.E$$

$$\bar{X} \pm Z_{\alpha/2} \frac{\sigma}{\sqrt{n}}$$

Tabl value

$\mu_{os} = 5$

$Z = 1.65$

$$5.37 \pm 1.65 \left(\frac{1}{\sqrt{200}} \right)$$

$$(5.25 \text{ to } 5.48)$$

$$P(5.25 \leq \mu \leq 5.48) = 0.95$$

Since Z is $>$ than Z_{α} reject H_0 and
~~and~~ conclude that mean sales transaction
 per customer has increases from the
 current level of 5\$ due to shont
 verbal communication.

Number 1

$$\begin{aligned}
 i) \quad n_1 &= 125 \\
 \bar{X}_1 &= 1.91 \\
 s_1 &= 1.32 \\
 \alpha &= 5\%
 \end{aligned}$$

$$\begin{aligned}
 n_2 &= 86 \\
 \bar{X}_2 &= 0.21 \\
 s_2 &= 0.53
 \end{aligned}$$

i) NM $H_0: \mu_1 = \mu_2$ ie $\mu_1 - \mu_2 = 0$
 AM $H_A: \mu_1 > \mu_2$
 ie the mean no of job changes is higher for British entrepreneurs than for British Corporate managers.

ii) $\alpha = 5\%$

$$\begin{aligned}
 iii) \quad TS &= \frac{(\bar{X}_1 - \bar{X}_2) - (\mu_1 - \mu_2)}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}} \rightarrow t_{(n_1+n_2-2)} \text{ df} \\
 &= \frac{1.7}{0.1312} = 12.96
 \end{aligned}$$

iv) tab val
 $t_{\alpha} (0.05, 209 \text{ df})$
 $= 1.645$

Inference: Since $T_{is} > T_{\alpha}$ reject null hypothesis and conclude that there is evidence to prove that mean number of job changes of BE are higher than British Corporate managers.